

Assignment 2

This assignment requires the implementation and comparative evaluation of recent models using appropriate datasets or simulation environments. The work must include relevant model selection, consistent experimental settings, suitable evaluation metrics, clear result analysis, and complete submission of code, data sources, and reproducibility details. The selected research area must remain consistent with Assignment 1, and all models and experiments must be directly related to that area.

Requirements

One of the following experimental settings must be adopted:

- 1. Dataset-Based Evaluation:** Implement and evaluate **4–6 recent models** on **2–3 relevant datasets** associated with the selected research problem.
- 2. Combined-Dataset Evaluation:** Merge **2–3 compatible datasets** to construct a new experimental dataset and evaluate **4–6 recent models**. The dataset sources, preprocessing, label harmonization, class distribution, and train/validation/test split must be clearly documented.
- 3. Simulator-Based Evaluation:** For research conducted in a simulation environment, implement and compare **3–4 recently proposed models** using the same or comparable simulation settings.

Model Selection

The selected models must be **directly and strongly related to Assignment 1**.

Models may be selected from:

- articles reviewed in Assignment 1;
- additional recent publications addressing the same or a closely related research problem;
- official implementations released by the original authors;
- reliable open-source repositories;
- independently developed implementations;
- implementations developed with assistance from AI-based coding tools.

Important: Models that are not sufficiently related to the research direction established in Assignment 1 will **not be considered for evaluation**.

Code and Implementation

Existing implementations from GitHub, Hugging Face, CodeShare, official project repositories, or other credible sources may be used.

AI tools may also be used for implementation support, debugging, preprocessing, code modification, and documentation.

However, the submitted implementation must be fully understood and verified. The model architecture, training procedure, loss function, major hyperparameters, input/output structure, and experimental workflow must be explainable.

All externally obtained code, pretrained models, repositories, and other resources must be properly acknowledged.

Experimental Setup

The experimental configuration must be clearly reported, including, where applicable:

- dataset or simulator name;
- data source;
- preprocessing procedure;
- train/validation/test split;
- input size;
- batch size;
- number of training epochs;
- optimizer;
- learning rate;
- loss function;
- augmentation strategy;
- pretrained weights;
- hardware and software environment.

Comparable experimental conditions should be maintained across models whenever possible to ensure a fair evaluation.

Evaluation and Results

Appropriate performance metrics must be selected according to the research problem.

Classification:

Accuracy, Precision, Recall, $F1$ -score, AUC

Segmentation:

Dice, IoU

Regression or Forecasting:

MAE, RMSE, R^2

For simulator-based studies, appropriate reward, success, safety, convergence, or other task-specific metrics must be reported.

The final results must include a clear comparative evaluation of all selected models. Numerical results should be supported by relevant tables, figures, curves, confusion matrices, qualitative outputs, or simulation results where applicable.

Submission Requirements

The final submission must include:

- a formal technical report;
- complete source code;
- dataset files, where redistribution is permitted, or official dataset links;
- merged-dataset preparation code, if applicable;
- experimental configuration files;
- result tables and figures;
- training/testing logs where relevant;
- references to all selected articles;
- links or citations for external code repositories;
- a README file containing clear instructions for reproducing the experiments.

A change to an unrelated research topic, unrelated models, or an unrelated experimental problem will not satisfy the requirements of this assignment.